The same nP_r can be written in factorial notation as follows:

$$nP_r = \frac{n!}{(n-r)!}$$

For example to find $10\ P_3$, we write in

factorial notation as $10\ P_3 = \frac{10!}{(10-3)!}$

$$10\ P_3 = \frac{10!}{7!} = \frac{10 \times 9 \times 8 \times (7!)}{(7!)}$$

$$= 10 \times 9 \times 8$$

$$= 720$$

Also we get the value for $10P_3$ as follows:

$$10\ P_3 = 10 \times (10\text{-}1) \times (10\text{-}2)$$

$$= 10 \times 9 \times 8$$

$$= 720$$

[To find $10\ P_3$, Start with 10, write the product of 3 consecutive natural numbers in the descending order]

Example 7.5

In how many ways can five students stand for a photograph in a row?

Solution:

The number of ways in which 5 students can stand in a row is same as the number of arrangements of 5 different things taken all at a time.

NOTE

(1) Number of all permutations of n things taken all at a time is n!

(2) $n\ P_0 = 1$, $n\ P_1 = n$ and $n\ P_n$ =n!

This can be done in $5P_5$ ways and $5P_5 = 5! = 120$ ways.

Permutation of objects not all distinct:

The number of permutations of n objects, where p_1 objects are of one kind, p_2 are of second kind … p_k are of kth kind and the rest, if any, are of different kind is given by

$$\frac{n_!}{p_1!p_2!....p_k!}$$

Example 7.6

Find the number of permutations of the letters in the word 'STATISTICS'